

Model Performance Test

Data Analytics

Project Name: AI-Enhanced Intrusion Detection System

Made by

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Model Performance Testing

Project team has filled the following information in the model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Dashboard Design	No. of Visualizations/Graphs – 4 (Live Threat Log, Threat Distribution Chart, Attack Type Breakdown, System Health Monitor)
2.	Data Responsiveness	Dashboard is fully responsive with real-time packet flow simulation. Slate UI components update dynamically as new flows arrive.
3.	Amount of Data Rendered (DB2 Metrics)	CICIDS2017 Dataset – 2,830,743 raw flow records; balanced subset: ~10,000 rows with SMOTE augmentation. Model serves predictions under 1ms per request.
4.	Utilization of Data Filters	On-Demand Scanner with 4 input parameters (Flow Duration, Total Fwd Packets, Total Bwd Packets, Total Length of Fwd Packets). Quick Test Presets: Benign, Brute Force, SQL Injection, XSS.
5.	Effective User Story	No. of Scenes Added – 3 (Dashboard Overview, On-Demand Scanner, Developer API Docs)
6.	Descriptive Reports	No. of Visualizations/Graphs – 4 (Threat log table, prediction confidence breakdown, feature value display, analyst security report card)

Dashboard Highlights

- ▶ 4 real-time visualizations covering live threats, distribution, attack breakdown, and system health
- ▶ On-demand scanner with preset profiles for quick testing across all 4 attack categories
- ▶ Responsive Slate UI that dynamically updates as new packet flows arrive
- ▶ 3 functional scenes: Dashboard Overview, On-Demand Scanner, Developer API Docs

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